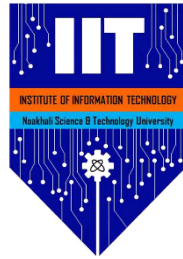


Institute of Information Technology, Noakhali Science and Technology University

Bachelor of Science in Software Engineering

Course Code: SE 4100



Internship Report

Submitted by

Md Masum Billah

ASH2125033M

Submitted to

Internship Placement Office

Institute of Information Technology (IIT)
Noakhali Science and Technology University

Performed at

B-Trac Solutions Ltd

Plot-68, Block-H, Road-11

Banani, Dhaka-1213, Bangladesh



Internship Attended: September 07, 2025 – January 29, 2026

Date of Submission: January 29, 2026



LETTER OF TRANSMITTAL

January 29, 2026

The Chairman

Internship Placement Office
Institute of Information Technology (IIT)
Noakhali Science and Technology University

Subject: Letter of Transmittal

Dear Sir,

It is a pleasure to submit the “Internship Report” as per the Internship Program requirement for the course SE 4100 Internship.

This report contains gory details of my activities during the Internship period. I have been working as an intern at B-Trac Solutions Ltd. I was supervised by Md. Mazba Kamal. I hope this report fulfils the requirements of the Internship Program.

I am requesting that you accept and consider this report.

Yours sincerely

Md Masum Billah

ASH2125033M

Session: 2020-21

BSc. in Software Engineering Program

Institute of Information Technology (IIT)

Noakhali Science and Technology University



DECLARATION

TO WHOM IT MAY CONCERN

I, **Md Masum Billah**, bearing ID: ASH2125033M, BSc. in Software Engineering Program, hereby declare that the presented “Internship Report” is uniquely prepared by me after completion of six months’ work in **B-Trac Solutions Ltd.**

My original work is submitted to the Institute of Information Technology (IIT), Noakhali Science and Technology University, and no part of the report has been submitted for any other degree or fellowship, and the work has not been published in any journal or magazine.

Md Masum Billah

ASH2125033M

Session: 2020-21

BSc. in Software Engineering Program

Institute of Information Technology (IIT)

Noakhali Science and Technology University



LETTER OF ENDORSEMENT

TO WHOM IT MAY CONCERN

This is to certify that **Md Masum Billah** was an intern at **B-Trac Solutions Ltd.** During this period, I was one of his supervisors.

I hereby confirm that I have reviewed the entire report. The contents of this report are true and not confidential to the company. The projects and training assignments mentioned in this report had the successful participation of Md Masum Billah.

I wish him all the very best for his future life.

Md. Mazba Kamal

Development Lead
B-Trac Solutions Ltd

Safayet Abdullah

Head of Projects
B-Trac Solutions Ltd



CERTIFICATE OF APPROVAL

This Internship report, submitted by **Md Masum Billah, ID No: ASH2125033M**, to the Chairman of the Internship Placement Office, Institute of Information Technology (IIT), Noakhali Science and Technology University, has been accepted as satisfactory for the partial fulfilment of the requirements for the degree of Bachelor of Science in Software Engineering and approved as to its style and contents. The presentation was held in January 2026.

Internship Examination Committee:

Mohammed Nizam Uddin, Chairman

Associate Professor & Director (Acting)
Institute of Information Technology (IIT)
Noakhali Science and Technology University

Md. Iftekharul Alam Efat, Committee

Member

Assistant Professor
Institute of Information Technology (IIT)
Noakhali Science and Technology University

Md. Eusha Kadir, Committee Member

Assistant Professor
Institute of Information Technology (IIT)
Noakhali Science and Technology University

Tasnim Rahman, Committee Member

Assistant Professor
Institute of Information Technology (IIT)
Noakhali Science and Technology University



DEDICATION

I dedicate this internship report to my parents, whose unconditional love, sacrifices, and continuous encouragement have been the foundation of my academic and personal development. Their constant support and belief in my abilities have motivated me to pursue my education with dedication and perseverance.

This work is a reflection of their guidance and patience, which have inspired me to overcome challenges and strive for excellence. I am deeply grateful for their role in shaping my values, discipline, and commitment toward achieving my professional goals.

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OFFER LETTER



Date: September 24, 2025

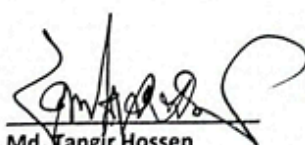
To
Director,
Career and Placement Center (CPC),
Noakhali Science and Technology University (NSTU)
Bangladesh.

Subject: Confirming letter of Internship.

Dear Sir / Madam,

We would like to inform you that Md. Masum Billah, ID no: ASH2125033M, Session: 7th Semester, contact no: 01400095352 has started his internship program with B-Trac Solutions Ltd. in the Web Applications department. The duration of the internship program is six months.

Thank you,



Md. Tangir Hossen
Strategic HR Partner



B-Trac Solutions Ltd.

Plot-68, Block-H, Road-11, Banani, Dhaka-1213, Bangladesh

+8802-883 8001-4 +8802-883 6868 info@btracsl.com

www.btracsolutions.com



ACKNOWLEDGEMENT

I would like to express my sincere gratitude to B-Trac Solutions Ltd for providing me with the opportunity to complete my internship in a professional and supportive environment. The organization offered an excellent platform to apply theoretical knowledge to real-world projects, enabling me to gain valuable industry exposure and practical experience in software engineering.

I am deeply thankful to Md. Mazba Kamal, Development Lead (Web), who served as my mentor throughout the internship period. His continuous guidance, technical insights, and constructive feedback significantly enhanced my learning experience. His mentorship not only strengthened my technical skills but also helped me understand professional responsibilities and industry standards.

I am also deeply thankful to Md. Iftekharul Alam Efat, Assistant Professor, Institute of Information Technology (IIT), Noakhali Science and Technology University (NSTU), for his academic supervision and encouragement. His guidance and support were instrumental in connecting academic learning with practical industry experience and in successfully completing this internship report.



PREFACE

This internship report has been prepared as a partial fulfillment of the academic requirements of my undergraduate program. The internship represents an essential phase of my academic journey, allowing me to apply theoretical knowledge gained throughout my coursework to real-world professional practice. The report documents my experiences, responsibilities, learning outcomes, and overall professional development during my internship period.

I undertook my internship at B-Trac Solutions Ltd, where I worked as a Software Engineer Intern. The primary purpose of this internship was to gain practical exposure to the software development industry and to understand how academic concepts are implemented in professional environments. Through hands-on involvement in real projects, I was able to observe industry workflows, development methodologies, and professional standards followed in a technology-driven organization.

During the internship, I was engaged in several development activities that contributed meaningfully to ongoing organizational projects. These tasks provided valuable insights into software engineering practices, including design principles, development, deployment, and collaboration within a multidisciplinary team. The experience also presented challenges such as adapting to new technologies, understanding complex systems, and meeting professional expectations, all of which played a vital role in my learning process.

EXECUTIVE SUMMARY

This internship report presents a comprehensive overview of my internship experience as a Software Engineer Intern at B-Trac Solutions Ltd. The internship was conducted as part of the academic requirements of my undergraduate program. The report outlines the organization profile, nature of work, responsibilities undertaken, learning outcomes, and overall professional development achieved during the internship period.

The primary objective of the internship was to bridge the gap between academic learning and practical application. Throughout the internship, I worked on real-world software development projects that required applying concepts such as web technologies, API integration, and software engineering best practices. The internship also aimed to develop professional skills such as teamwork, communication, time management, and problem-solving.

To achieve these objectives, a practical and experiential methodology was followed. The internship involved active participation in project development under the supervision of an experienced mentor. Learning was facilitated through hands-on tasks, regular discussions, code reviews, and collaboration with team members. Additional information was gathered through organizational documentation, official company resources, and interactions with colleagues and fellow interns.

The major outcomes of the internship include improved technical proficiency in modern software development technologies, enhanced understanding of industry workflows, and increased confidence in handling professional responsibilities. Working on production-level systems and real-time applications significantly strengthened my analytical and problem-solving abilities.

In conclusion, the internship at B-Trac Solutions Ltd was a valuable learning experience that contributed significantly to my technical and professional growth. The knowledge and skills gained during this period have prepared me for future academic endeavors and a professional



career in software engineering. The report concludes that such industry-based internships are essential for developing competent and industry-ready graduates.

Introduction

Preamble

From the first semester of my undergraduate program, the academic journey has been centered on building a strong foundation in software engineering. Early semesters focused primarily on theoretical concepts such as structure programming, data structures, algorithms, object oriented programming, databases, and design patterns. As I progressed through subsequent semesters, these concepts were reinforced through laboratory work, coursework, and project-based learning, allowing me to gradually develop practical and technical skills alongside academic knowledge.

Over the course of six semesters, continuous guidance from faculty members, structured curricula, and academic assessments helped shape my analytical thinking, problem-solving ability, and understanding of engineering discipline. However, despite gaining theoretical knowledge and completing academic projects, it became evident that real-world software development environments involve complexities that cannot be fully captured within classroom settings alone.

The academic internship represents the final and most critical phase of this learning journey. It serves as a bridge between university education and professional practice by allowing students to apply their accumulated knowledge in an industry setting. This internship has played a significant role in preparing me for my future career by exposing me to real-world challenges, professional responsibilities, and collaborative work culture. As an academic requirement, the internship not only complements theoretical education but also strengthens employability and career readiness by aligning academic learning with industry expectations.

Origin of the report

This report has been prepared as part of the mandatory academic internship requirement of my undergraduate program. The purpose of this report is to formally document my internship journey, including the nature of the organization, assigned responsibilities, learning outcomes, and professional experiences gained during the internship period. It serves as an academic record that reflects how the internship contributed to my personal, technical, and professional development and fulfills the reporting requirement set by the faculty.

Objective of the internship

The internship was intended to provide practical exposure to the software industry, enhance technical competency, and develop professional skills such as teamwork, communication, and responsibility. Another key objective was to understand workplace ethics, organizational structure, and real-world problem-solving approaches, which are essential for a smooth transition from academic life to a professional career.

Scope

The scope of the internship extended beyond technical skill development. It provided an opportunity to apply academic knowledge to real-world problem implementation, work collaboratively within a team, and understand the connection between theory-based education and practical software development. Through this internship, I was able to experience industry-standard workflows, tools, and development practices. Additionally, the internship opened pathways for professional networking and provided valuable exposure that may contribute to future employment opportunities within the industry.

Methodology

This report has been prepared based on a qualitative and experiential methodology. The primary foundation of the report is my personal experience within the organization during the internship period. Information was gathered through daily work activities, direct interaction with mentors, and participation in team discussions.

Additional information was collected from organizational resources such as the company's official website, LinkedIn page, and Facebook presence to better understand its background, culture, and professional practices. This combination of personal experience, peer interaction, mentorship guidance, and organizational resources forms the basis of the content presented in this report.

Limitations

Despite the valuable learning opportunities provided by the internship, certain limitations were inherent due to my role as an intern. As an intern, my involvement was primarily focused on assigned tasks and supervised project work, and I was not included in all aspects of the organization's core operational and decision-making processes. Consequently, some internal workflows, strategic discussions, and high-level project planning activities remained beyond my scope of exposure.

Additionally, due to confidentiality and data security policies, access to sensitive information, proprietary systems, and critical business data was restricted. As a result, certain technical, managerial, or operational details could not be observed or documented in this report. Therefore, the observations and analyses presented are based solely on the areas and responsibilities to which I was formally exposed during the internship period.

Recruitment

Overview

The recruitment process for the internship was established through a strong academic–industry collaboration between our institute and several reputed technology companies. Over the years, previous batches of students had completed internships and professional placements through these partnerships, which helped build mutual trust between the academy and industry partners. Based on this ongoing collaboration, companies agreed to consider students from our institute for internship opportunities as part of the academic internship requirement. This structured partnership ensured that the recruitment process aligned with academic objectives while meeting industry expectations.

On-Site Interview

As part of the recruitment process, student resumes were forwarded by the faculty to multiple collaborating organizations, including Brain Station 23, Samsung R&D, Brac IT Services, and other technology firms. Following the initial resume screening, shortlisted candidates were invited to participate in company-specific assessments and interview processes, conducted according to each organization’s recruitment policies.

These evaluations typically included technical interviews, problem-solving discussions, and behavioral assessments. Participating in such interviews provided valuable exposure to real-world hiring practices and helped strengthen the professional relationship between the academy and industry partners. This experience also had a positive impact on career readiness, as it familiarized students with industry expectations before graduation.

On-Campus Recruitment

As part of the on-site recruitment process, initially, nine companies expressed interest in participating, however, at the final stage, three organizations conducted recruitment activities on campus: B-Trac Solutions Ltd, OS IT, and Acquant Technologies. This initiative marked a significant milestone for the institute, strengthening its academic reputation and enhancing employment opportunities for students.

OS IT primarily focuses on remote internship opportunities and conducts an interview session on campus. Selected students from the faculty participated in this session, which was attended by the company’s CEO, Mohammad Moniruzzaman, and Director, Nusrat Jahan.

Acquant Technologies arranged an interview session followed by a task-based assessment. Selected candidates were assigned a development task, and final evaluation was conducted based on task performance. The session was attended by Mahabub Alam Tito (CTO), Mahafuzur Rahman (COO), and Robiul Awal (Web & Digital Strategist).

B-Trac Solutions Ltd followed a unique recruitment approach that emphasized creativity, communication, and mindset rather than traditional technical screening alone. Instead of conventional interviews, the company arranged a group discussion session. The discussion

was led by Mohammed Tanvir Siddique (CEO), who interacted directly with students to assess their academic journey, thought process, motivation, and career vision.

Personal Experience

I had the opportunity to participate in multiple interview sessions during this recruitment phase. I attended an interview at Samsung R&D, where questions were primarily based on my resume, completed projects, and technical knowledge related to Laravel, as the position targeted that domain. I was also asked about my extracurricular activities and competitive programming experience, which helped me understand the importance of a well-rounded profile.

Additionally, my resume was forwarded to Webase Technologies, where I participated in a short online interview session lasting approximately 10–15 minutes. The interview focused mainly on behavioral questions and Laravel-related concepts, as the organization was seeking candidates with experience in that area.

Through the on-campus recruitment program, I also participated in the interview process with Acquant Technologies, where the discussion was largely behavioral, followed by task-based evaluation.

Most notably, I participated in the group discussion session with the CEO of B-Trac Solutions Ltd. The discussion covered a wide range of topics, including our academic learning, practical skill development, potential contributions to the organization, career goals, expectations from interns, and the ongoing projects within the company. This interactive and open discussion provided deep insight into professional thinking and organizational vision.

Final Selection and Appointment

After completing both on-campus and off-campus recruitment processes, I was finally selected as a Software Engineer Intern at B-Trac Solutions Ltd. I received a formal phone call from the HR representative, Afsana Akther, informing me of my selection. She subsequently shared the joining details through SMS. This selection marked a significant achievement in my academic journey and represented the successful transition from university learning to professional industry exposure.

Parent Company

Overview of Bangla Trac Group

Bangla Trac Group is a leading and diversified conglomerate in Bangladesh that has been operating for over eighteen years, making a significant impact on the national economy and key industrial sectors. The group first began its journey in 2007 with the launch of Bangla Trac Communications as a licensed International Gateway (IGW) service provider, and it has since evolved into a multi-sector business organization with interests across technology, energy, infrastructure, and retail markets.

The group's contribution to the country has been substantial, especially in power generation, telecommunications, information technology, and infrastructure development. It has supported national development objectives by delivering reliable services across critical sectors, contributing to economic growth and technological advancement. The group's operations emphasize quality, innovation, and customer-centric services, which have helped strengthen Bangladesh's industrial capacity and service delivery.

Bangla Trac Group currently operates through multiple business units that engage in power generation, IT and digital solutions, telecommunications, and international retail franchises. Its strategic vision focuses on sustainable growth, technological integration, and the establishment of globally recognized brands within the national market. The group continues to expand its footprint while reinforcing its commitment to economic development and service excellence.

Business Area

Bangla Trac Group's business operations are diversified across several major sectors that contribute to both national infrastructure and service innovation.

- **Power and Infrastructure:** Through its power generation units, the group contributes significant megawatt capacity to the national grid, supporting energy supply and infrastructure growth.
- **Information Technology and Digital Services:** The group delivers IT solutions, digital services, and software and hardware offerings that support both private industry and government projects.
- **Telecommunications and Connectivity:** Through its telecommunications ventures, the group provides international gateway services, data hosting, and network solutions that enhance connectivity within and beyond Bangladesh.
- **Retail and Franchise Services:** The group operates international retail franchises, including major global brands, contributing to the local economy through service, employment, and consumer engagement.

Each of these sectors aligns with the group's overarching objective of contributing to Bangladesh's socio-economic progress through diversified yet integrated business activities.

Sister Company

Bangla Trac Group comprises several affiliated and subsidiary companies, each specializing in distinct industries and contributing to the group's diversified business portfolio.



Figure 01: Sister Company of Bangla Trac Group

B-Trac Solutions Ltd: A technology services company focused on software development, digital solutions, and IT consulting.

B-Trac Technologies Ltd: A value-added IT distribution company that manages partnerships with leading global technology brands.

Bangla Trac Communications Ltd: A pioneering telecommunication and international gateway service provider.

Bangla Trac Engineering Ltd: A company specializing in engineering solutions, manufacturing, and support services for industrial and power systems.

Together, these companies contribute to the group's expansive presence across technology, power, engineering, and service industries, reinforcing its role as a major corporate force in Bangladesh.

Partner Company

Bangla Trac Group maintains partnerships with a number of respected global and local organizations that enhance its operational reach and service capabilities.



Figure 02: Partner Company of Bangla Trac Group

Leading Technology Vendors: Through affiliated companies such as B-Trac Technologies Ltd., the group partners with global technology brands including Dell, HP, Microsoft, Panasonic, AVITA, and BlackBox, supporting digital transformation and IT distribution in Bangladesh.

Telecommunications and Data Service Entities: Collaborative relationships in telecommunications and international gateway services help the group provide robust connectivity and data solutions.

These partnerships reinforce the group's ability to deliver comprehensive solutions and maintain high standards of service across multiple sectors.

Company Profile

Overview of B Trac Solutions

B-Trac Solutions Ltd is a technology company based in Dhaka, Bangladesh, specializing in the design, development, and implementation of software and application systems across web, mobile, and IoT domains. Established as a core IT enterprise under Bangla Trac Group, the company contributes to digital transformation by delivering innovative software solutions to government agencies, corporate organizations, and enterprise clients.

The company's operations have had a positive impact on the Bangladesh IT industry by driving adoption of locally developed digital systems that address real-world business and social challenges. Its solutions emphasize user-centric design, cloud integration, and adaptive technologies that support data-driven decision-making and efficiency improvements in public and private sector workflows.

B-Trac Solutions also strengthens the broader Bangla Trac Group by adding a technology and digital services arm to the conglomerate's diversified portfolio. This integration helps the group offer end-to-end solutions that combine software innovation with existing infrastructure, telecommunications, and enterprise services, thereby supporting the national economy and enhancing industry standards.

Mission and Vision

Vision: To become a dynamic global technology company that drives stakeholder success through excellence, integrity, humility, and people-centric leadership.

Mission: To add value and transform user experiences by addressing real-life challenges through technology and innovation.

Core Values and Strategic Focus

B-Trac Solutions Ltd operates based on a set of core principles that guide its development processes and client relationships:

- **Innovation:** Continuous innovation fuels the organization's drive to deliver transformative solutions.
- **Efficiency:** Automation and optimized workflows are central to improving operational performance.
- **Reliability:** A strong commitment to delivering dependable and seamless solutions.
- **Customer Centricity:** Emphasis on understanding and exceeding client expectations.
- **Collaboration:** Strategic partnerships and teamwork enable sustainable growth and long-term impact.

Company Hierarchy

The organizational structure of B-Trac Solutions Ltd is designed to support effective leadership, accountability, and technical excellence. The company is led by Mohammed

Tanvir Siddique, Chief Executive Officer (CEO), followed by senior management roles including heads of projects, research and development, sales, IT, and corporate finance.

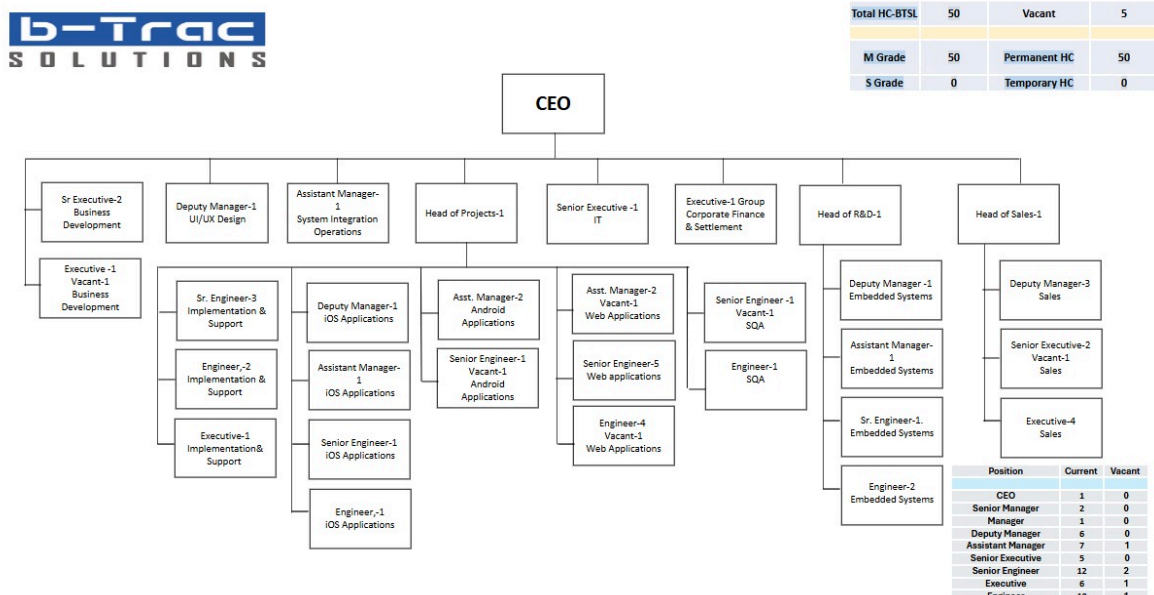


Figure 03: Hierarchy of B-Trac Solutions Ltd

Under these leadership roles, the organization is divided into specialized technical teams such as Web Applications, Android Applications, iOS Applications, Embedded Systems, Software Quality Assurance (SQA), System Integration, and Support Services.

Services

B-Trac Solutions Ltd provides a comprehensive range of technology services aimed at delivering end-to-end digital solutions:

- Custom Web Application Development
- Mobile Application Development (Android and iOS)
- System Integration services
- Cloud Services and infrastructure support
- IT Consultancy and digital transformation advisory
- Software implementation and technical support services

These services enable organizations to modernize operations, enhance productivity, and adopt scalable digital platforms.

Products

The company has developed and deployed several digital products designed to address real-world challenges:



Figure 04: Products of B-Trac Solutions Ltd

- **Carcopolo** – Vehicle safety and tracking solution
- **Yes Parking** – Smart parking management system
- **Nirapath** – Personal safety and emergency response application
- **Workopolo** – Workforce and task management platform
- **Seemo** – Visual security and monitoring solution

Each product reflects the organization's emphasis on usability, reliability, and technological innovation.

Projects

B-Trac Solutions Ltd has successfully delivered numerous projects across web and mobile platforms. Notable projects include:



Figure 05: Projects of B-Trac Solutions Ltd

- **CIMS** – Web-based Citizen Information Management System
- **DMP Duty** – Mobile application for duty and task management
- **Banani Club** – Integrated website and mobile application
- **Doorstep** – Web application for service delivery

These projects demonstrate the company's capability to design and implement scalable, secure, and user-focused system

Partner Company

B-Trac Solutions Ltd collaborates with leading global technology providers to deliver high-quality IT solutions, system integration, and cybersecurity services. Key partners include Dell, HP, Barracuda, Red Hat, Teknowledge, Sirius International Holding, and Presight.

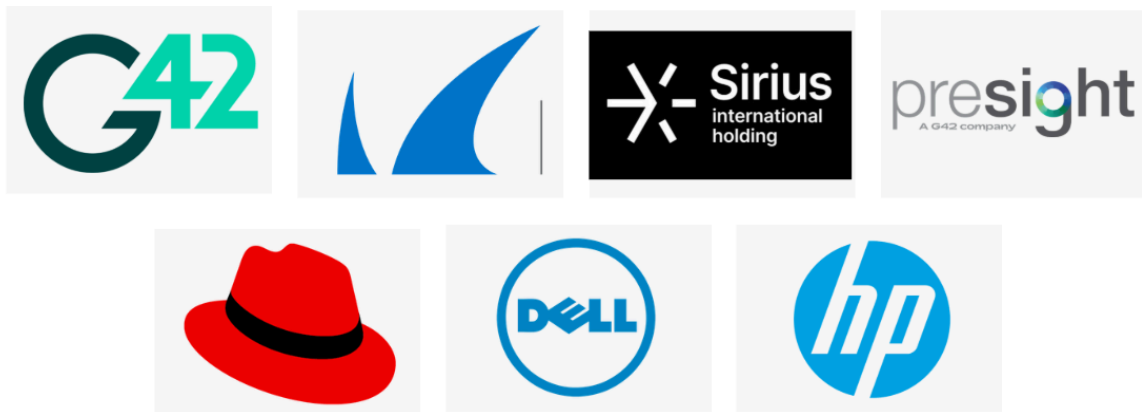


Figure 06: Partner Company of B-Trac Solutions Ltd.

These partnerships enable the company to provide enterprise-grade solutions that meet the evolving demands of government and corporate clients.

Awards and Participations

B-Trac Solutions Ltd has actively participated in national and international technology events and has received recognition for innovation and excellence:

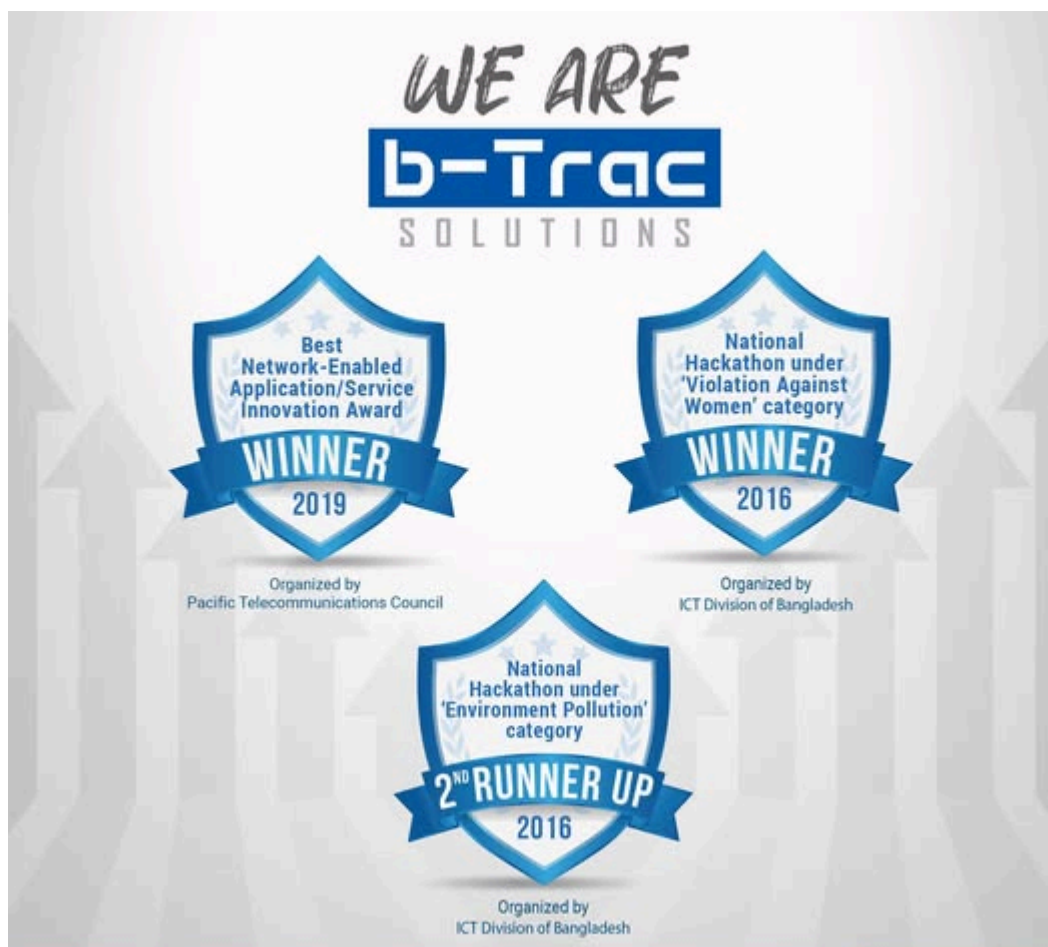
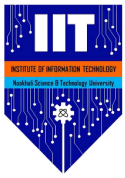


Figure 07: Awards of B-Trac Solutions Ltd.

- **National Hackathon – 2016 (Award)**
- **PTC Innovation – 2019 (Participant)**
- **Chittagong Motor Fest – 2022 (Participant)**
- **ITEX – 2023 (Award)**
- **Chittagong Motor Fest – 2024 (Participant)**
- **Dhaka Motor Fest – 2025 (Participant)**

These achievements highlight the company's commitment to continuous improvement and technological leadership.



Internship at B-Trac Solutions LTD

Overview

The internship at B-Trac Solutions Ltd was strongly aligned with my academic curriculum in software engineering. Fundamental concepts studied at the university such as structured programming language, web technologies, database management systems, software requirement specification (SRS), and software testing and quality assurance were directly applied during my internship tasks. This practical application reinforced my theoretical knowledge and enhanced my understanding through real-world problem-solving. As a result, the internship served as an effective bridge between academic learning and professional practice, improving both my technical competence and conceptual clarity.

Internship Duration

The internship at B-Trac Solutions Ltd was conducted over a period of five months, from 7 September to 29 December, in accordance with the academic requirements. During this period, the organization facilitated internship opportunities across multiple technical departments, including Embedded Systems, Android Development, Web Development, and other specialized software domains. This multi-departmental structure allowed interns to gain a comprehensive understanding of the organization's diverse technological operations.

Joining Day

On my joining day, I was instructed to report to the office at 10:00 A.M. Upon arrival, I reported to Afsana Akther (HR), who guided me through the necessary administrative procedures. During this session, I was formally introduced to the company's policies, workplace rules, and professional expectations, which provided a clear understanding of organizational standards and conduct.

After completing the administrative formalities, I was instructed to report to Md. Mazba Kamal (Development Lead, Web). He introduced me to the Development Team and provided an overview of the team structure, and my role as a Software Engineer Intern. This introduction helped me feel welcomed and prepared to begin my internship in a professional and structured manner.

Initial Guidance

During the initial phase of the internship, I received formal guidance from Safayet Ahmed (Head of Projects) regarding professional conduct, task execution, and workplace responsibilities. He provided clear instructions on organizational practices, project workflows, and expectations for interns, which helped me understand my role within the team and adapt to the professional environment.



Subsequently, Md. Mazba Kamal (Development Lead, Web) reviewed my resume and assessed my technical background. Based on this evaluation, I was assigned a full-stack development role, and he was officially designated as my mentor.

Roles and responsibilities

Working within the Web Development Team provided focused exposure to modern web technologies, development frameworks, and collaborative software engineering practices, thereby strengthening my practical skills and aligning closely with my academic background.

As a Software Engineer Intern, my assigned roles and responsibilities included:

- Designed and developed full-stack web applications following project requirements.
- Wrote clean, maintainable, and efficient code following coding standards.
- Debugged applications to ensure functionality, performance, and reliability.
- Supported testing activities to maintain application quality and stability.
- Deployed applications following standard deployment procedures.
- Documented development processes and maintained accurate records of project work.

Interns were expected to adhere to professional workflows, follow established coding standards, and maintain proper documentation. Over time, my responsibilities expanded to include independent problem-solving and active contributions to project modules, which enhanced both my technical proficiency and professional accountability.

Personal Growth

The internship at B-Trac Solutions Ltd contributed significantly to my personal and professional growth. Exposure to real-world software development challenges enhanced my problem-solving abilities, analytical thinking, and adaptability. Working within a professional development environment allowed me to better understand industry expectations and improve my ability to manage tasks effectively.

The team followed agile development principles, which provided me with the opportunity to observe and participate in iterative development cycles, sprint planning, and collaborative development processes. During the internship, I also had the opportunity to explore and work with modern technologies such as GraphQL APIs, GitHub GraphQL Public API, and Firebase Cloud Messaging (FCM). These experiences broadened my technical knowledge and strengthened my confidence in working with contemporary software development tools and architectures.

Work Environment

B-Trac Solutions Ltd maintains a professional, collaborative, and supportive work environment. Team members are approachable and actively encourage knowledge sharing

and teamwork. Supervisors regularly provide constructive feedback and technical guidance, which fosters continuous learning and continuous improvement.

The organization follows a structured yet flexible work culture, with standard office hours from 9:00 A.M. to 6:00 P.M. The workplace operates as a co-working environment with no fixed desk arrangements, promoting collaboration and effective communication among team members. The company observes Friday and Saturday as weekly holidays, and among the five working days, one day is allocated for remote (home office) work. This balanced and modern work environment contributes positively to productivity, work-life balance, and overall professional satisfaction.

Facilities

B-Trac Solutions Ltd provides adequate facilities to ensure a comfortable and productive work environment. The office offers unlimited coffee and lunch facilities, with lunch typically served after 2:00 PM. The workplace is designed to support collaboration, teamwork, and effective communication among employees. These facilities contribute to a positive organizational culture and enhance overall work efficiency.

Farewell

At the conclusion of the internship period, a formal farewell marked the completion of my training at B-Trac Solutions Ltd. During this time, feedback was provided regarding my performance, learning outcomes, and areas for future improvement. The farewell reflected the organization's appreciation for my dedication and contributions throughout the internship and served as a positive and motivating conclusion to the professional experience gained during this period.

Conclusion

The internship at B-Trac Solutions Ltd was a valuable and enriching experience that effectively combined academic knowledge with practical industry exposure. Through structured training, meaningful responsibilities, and a supportive work environment, the internship contributed significantly to my technical skill development and professional maturity. The experience and skills gained during this period will serve as a strong foundation for my future academic pursuits and professional career in software engineering.

Practice Projects

Taskify

Taskify is a command-line-based tool developed to manage Firebase projects locally. The tool enables users to register, inspect, and remove Firebase projects using service account JSON files. It incorporates validation mechanisms to ensure the correctness of credentials and maintains persistent local storage for efficient project management.

Motivation

As part of my internship responsibilities, I was assigned to work on an earthquake alert system intended to send real-time alerts to multiple mobile applications. To support this system, I was required to manage and maintain Firebase credentials for multiple applications efficiently.

Following this, I want to develop a solution that could handle Firebase service account credentials either through a graphical user interface or a command-line interface using *commander.js*. I chose to implement the solution as a command-line application, as this approach was new to me and provided an opportunity to explore a different development paradigm beyond traditional user interface-based systems.

Features

The key features of Taskify include:

- Interactive REPL mode (terminal stays alive)
- Non-interactive CLI commands
- Register Firebase projects using service account JSON
- Validate Firebase service account structure
- List all registered projects
- View details of a specific project
- Remove individual projects or clear all
- Persistent local storage (*projects.json*)
- Type-safe implementation with TypeScript
- Simple, predictable CLI interface

Technologies

- Node.js
- Commander.js
- TypeScript

Learnings

Through the development of this application, I gained practical experience in building command-line applications using Node.js and TypeScript. I also learned how to design a centralized logging system using *Winston*, implement input validation, and manage persistent local data storage. This project significantly enhanced my understanding of backend tooling, automation, and efficient project configuration management in real-world software systems.

Proof of Work

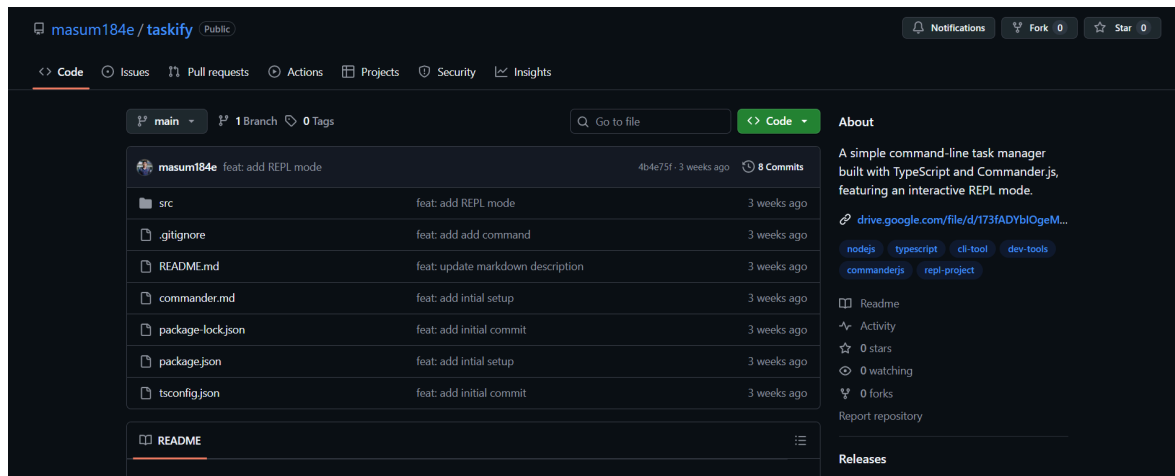


Figure 08: GitHub repository of Taskify

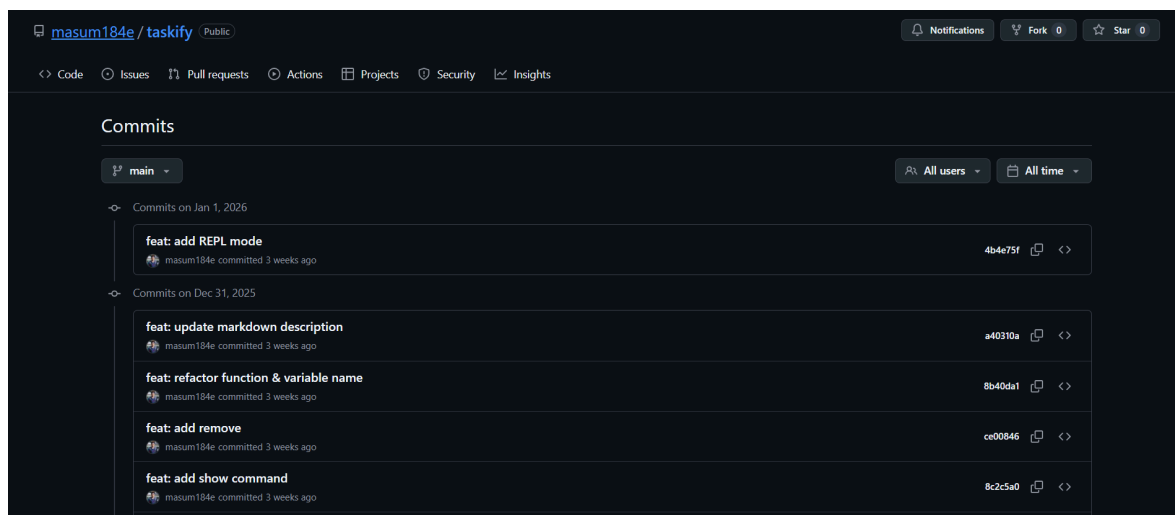


Figure 09: Commit history of Taskify

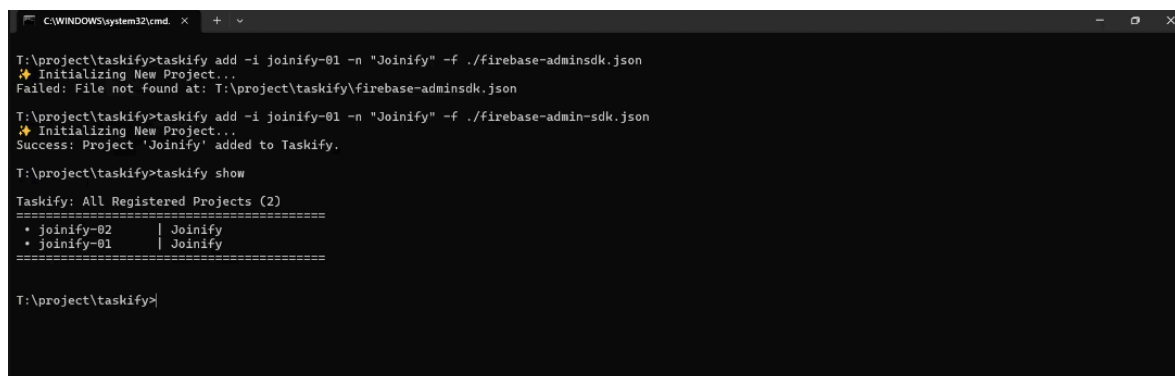


Figure 10: Command line interface of Taskify

GitHub Repository: <https://github.com/masum184e/taskify>

Video Preview: https://drive.google.com/file/d/173fADYbIOgeML2ZwCgnRUZ_rQLnJq4xk

FCM Playground

FCM Playground is an experimental sandbox application that I developed to explore and implement Firebase Cloud Messaging (FCM) using both the Firebase Client SDK and the Firebase Admin SDK. The application demonstrates the complete workflow of registering client devices, handling foreground and background notifications, and sending messages from a server-side environment. This project was designed to simulate real-world notification delivery scenarios in a controlled and testable setup.

Motivation

As part of my work on the earthquake alert system, I needed a reliable method to test the sending and receiving of alert messages using Firebase Cloud Messaging. I searched for existing mobile and web-based tools that could receive FCM notifications on devices and allow message transmission from a web interface. However, I could not find any solution that fully met my specific testing requirements.

As a result, I decided to design and develop my own testing platform. This approach allowed me to gain deeper technical insight into FCM while creating a reusable tool that could be applied directly to the earthquake alert system.

Features

Client SDK

- Smart input and validation of client credentials
- Receiving messages through the Firebase Client SDK
- Retrieval and display of the FCM device token
- Subscription to one or more messaging topics

Admin SDK

- Validation and configuration of the Firebase Admin SDK
- Sending messages to a specific device using its FCM token
- Sending messages to a group of devices based on topic subscription
- Broadcasting messages to all registered devices, with a default subscription topic (*all_users*)

Technologies

- Next.js

- Firebase
- Shaden UI
- Zod

Learning

By developing this application, I gained practical experience in implementing Firebase Cloud Messaging and managing real-time communication between servers and client devices. I developed a clear understanding of how FCM handles foreground and background notifications, as well as the role of the *firebase-messaging-sw.js* service worker in web-based messaging. Additionally, I learned how browser APIs such as the Navigator interface interact with service workers to enable real-time notifications. This project significantly strengthened my understanding of real-time messaging systems and event-driven communication architectures.

Proof of Work

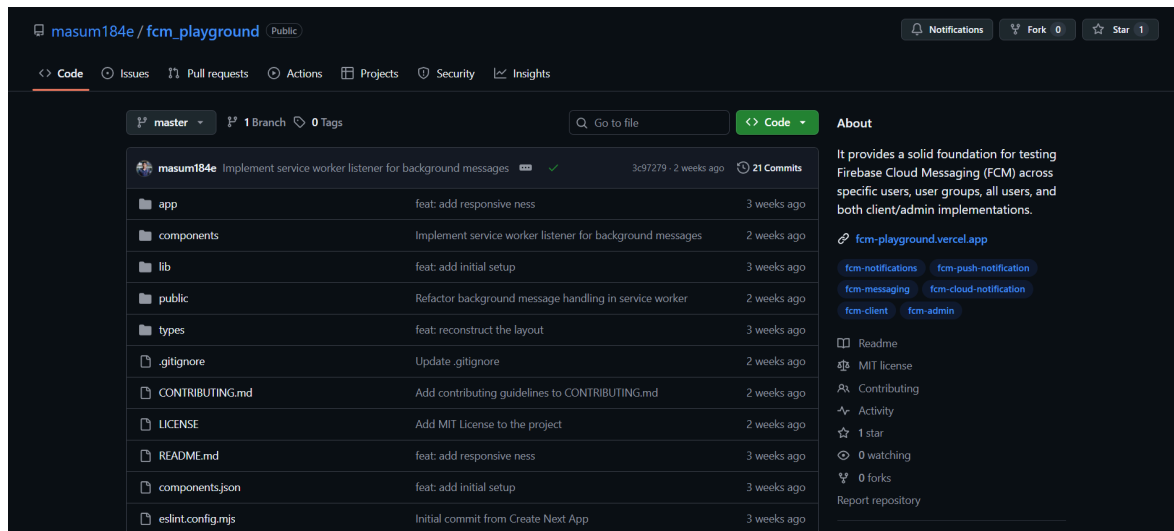


Figure 11: GitHub repository of FCM Playground

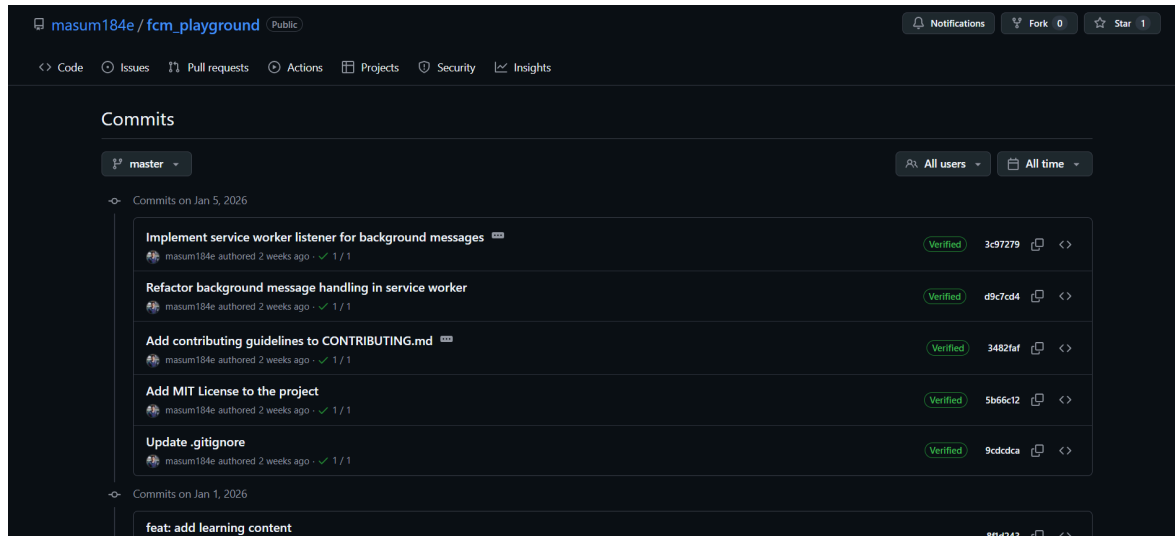


Figure 12: Commit history of FCM Playground

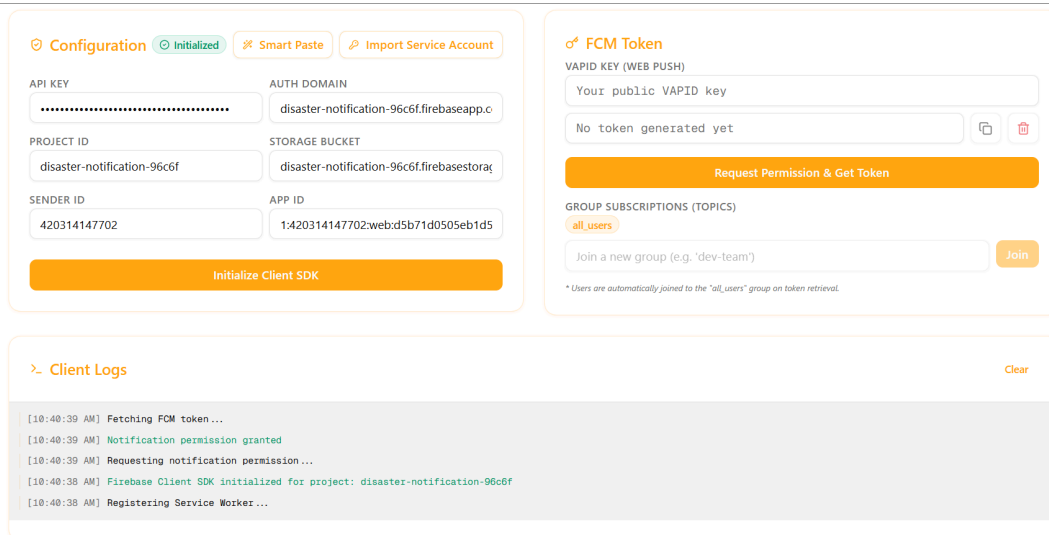


Figure 13: Client SDK interface for FCM Playground

Admin Configuration

Initialized

Paste your Service Account JSON from Firebase Console

SERVICE ACCOUNT JSON never persisted

```
{
  "type": "service_account",
  "project_id": "disaster-notification-96c6f",
  "private_key_id": "35d151ad7cb1cd9da22dac4371432eade72b10fe",
  "private_key": "-----BEGIN PRIVATE KEY-----
\\nMIEVQIBADANBgkqhkiG9w0BAQFAASCBKcwggSjAgEAAoIBAQCzQNi7yqdiEv2E\\nuNC
DwbMQU20XoYTSX2cZ0++e7URtg0QyUoQ2Y+/DCx5F5r0/3p11Fh2b0c+kyv4Yw\\n6XVWVuzT
CwWQF12yXrvZUmJf8wxWA1C8mboyCKoVjKtdvY7MrG30wEWtEI+n6dP/\\nBhHbcnnBfKzLN
-----",
  "client_email": "firebase-adminsdk-96c6f@disaster-notification-96c6f.iam.gserviceaccount.com",
  "auth_uri": "https://accounts.google.com/o/oauth2/auth",
  "token_uri": "https://oauth2.googleapis.com/token",
  "auth_provider_x509_cert_url": "https://www.googleapis.com/oauth2/v1/certs",
  "client_x509_cert_url": "https://www.googleapis.com/oauth2/v1/certs"
}
```

Validate & Initialize

Dispatch Message

Send a test notification to a specific client

☐ Specific ☐ Group ☐ All

REGISTRATION TOKEN

Paste token from Client column

TITLE BODY

Hello from FCM! This is a test notification.

Send to User

Server Logs

Clear

```
[10:45:28 AM] Admin SDK (Simulated) initialized for project: disaster-notification-96c6f
```

Figure 14: AdminSDK interface for FCM Playground

GitHub Repository: https://github.com/masum184e/fcm_playground

Live URL: <https://fcm-playground.vercel.app/>

Assigned Projects

BTSL GitHub Dashboard

The BTSL GitHub Dashboard is an analytical web-based tool that I worked on during my internship at B-Trac Solutions Ltd. The primary objective of this project was to enable team leaders to monitor and analyze the activity of team members within the B-Trac Solutions Ltd GitHub organization. The dashboard provides insights into individual and team-level contributions, supporting better project management and performance evaluation.

Features

The key features of the BTSL GitHub Dashboard include:

- Tracking recent activities across repositories, including commits, pull requests, and issues
- Analyzing pull request trends based on status (open, closed, merged), labels, and selected time periods
- Measuring issue activity, including open versus closed issues and label-based categorization
- Monitoring code contribution trends using commit history, additions, and deletions
- Identifying recently active repositories based on the latest push dates
- Profiling team members with detailed contribution metrics across commits, issues, and pull requests
- Evaluating team collaboration through pull request reviews and review timelines
- Supporting time-based reporting with flexible date range filters
- Enabling granular filtering by repository, contributor, label, and activity type for deeper analytical insights

Technologies

- Next.js
- GraphQL (GitHub Public GraphQL API)
- Docker
- Shadcn UI
- TypeScript

Learning

Although I had prior experience working with GraphQL, my earlier projects involved both frontend and backend development, where I had full control over the API and query structure. In contrast, this project required working with the GitHub Public GraphQL API, which presented new challenges in query design and data retrieval, as I was unfamiliar with how to construct the necessary queries.

Initially, I found it challenging to construct GraphQL queries that exactly matched the project requirements. After exploring various resources, I discovered the Apollo GraphQL API Explorer, which proved to be an effective tool for exploring the schema and building complex queries interactively. This experience significantly improved my ability to understand and work with public GraphQL APIs, extending beyond GitHub to other external APIs.

While developing the backend logic, I also encountered and learned to handle the pagination behavior of the GitHub GraphQL API, particularly when working with large volumes of data. Managing paginated responses improved my understanding of data fetching strategies and performance considerations.

Following development, I was responsible for deploying the application to a Virtual Private Server (VPS) using credentials provided by the organization. Although the application functioned correctly in the local environment, several issues arose during deployment. Initially, the entire user interface failed to load styles correctly due to a proxy configuration issue on the VPS, which I successfully identified and resolved.

Subsequently, I encountered inconsistencies in API behavior, where some requests succeeded while others failed. Through detailed debugging, I determined that the issue stemmed from differences between Next.js client-side and server-side components, specifically related to network request handling. Resolving these issues enhanced my understanding of Next.js client and server components, deployment environments, and real-world production debugging.

Proof of Work

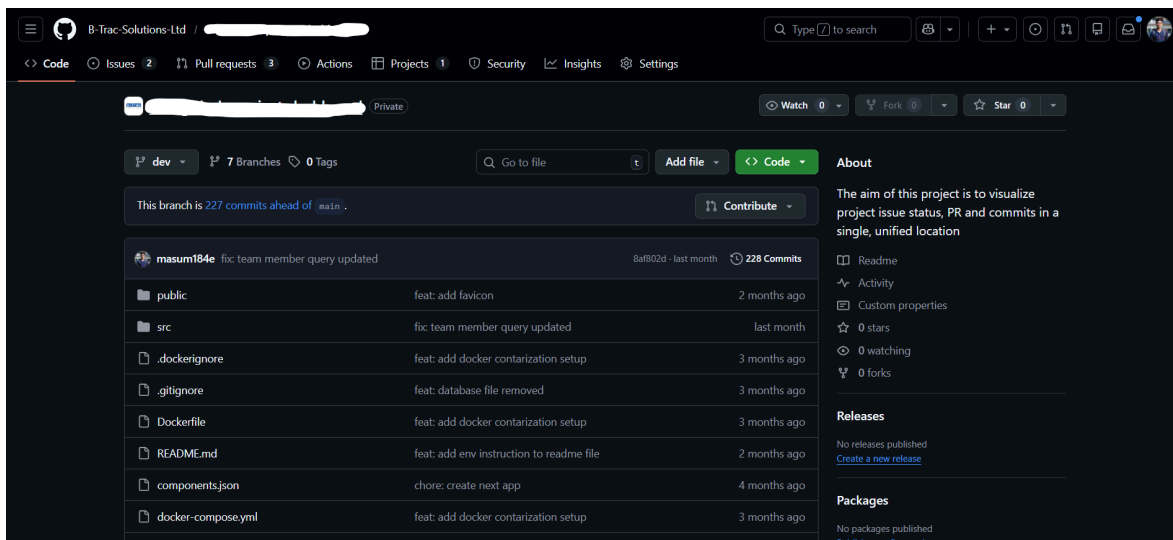


Figure 15: GitHub repository of GitHub dashboard

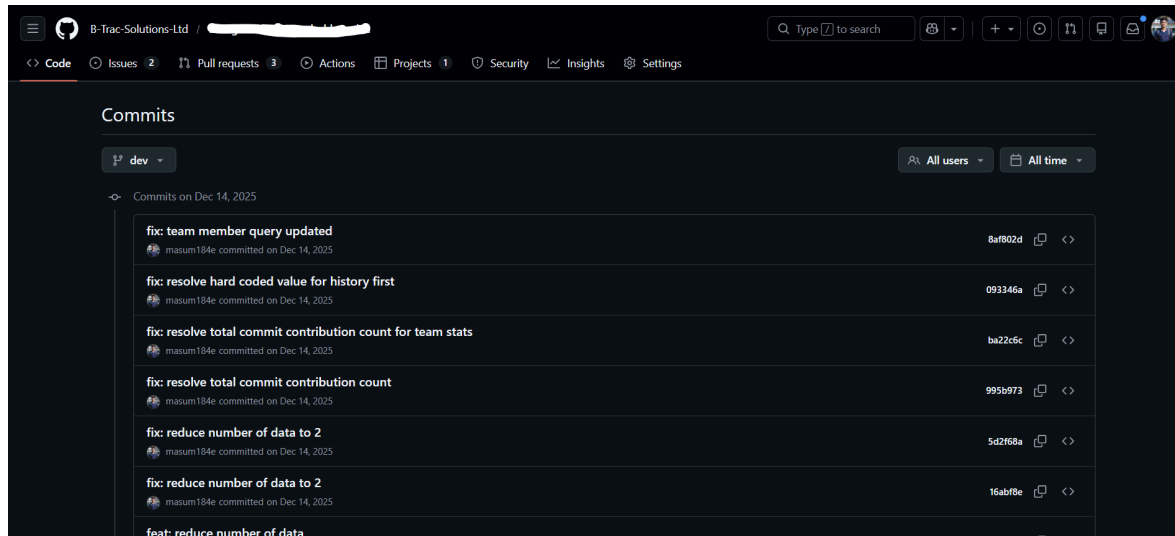


Figure 16: Commit history of GitHub dashboard

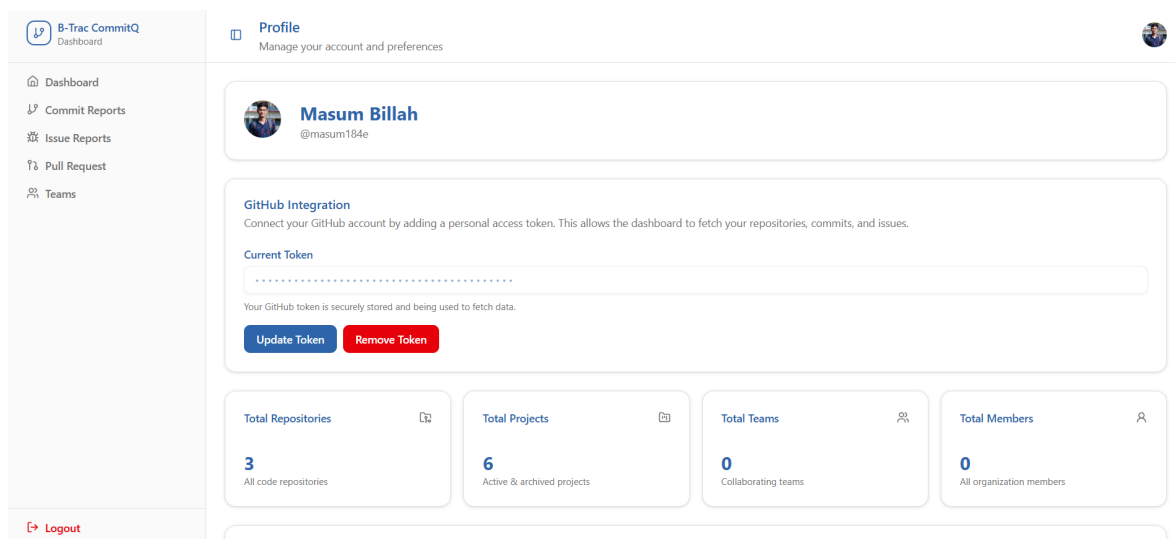


Figure 17: Team leader profile of GitHub dashboard

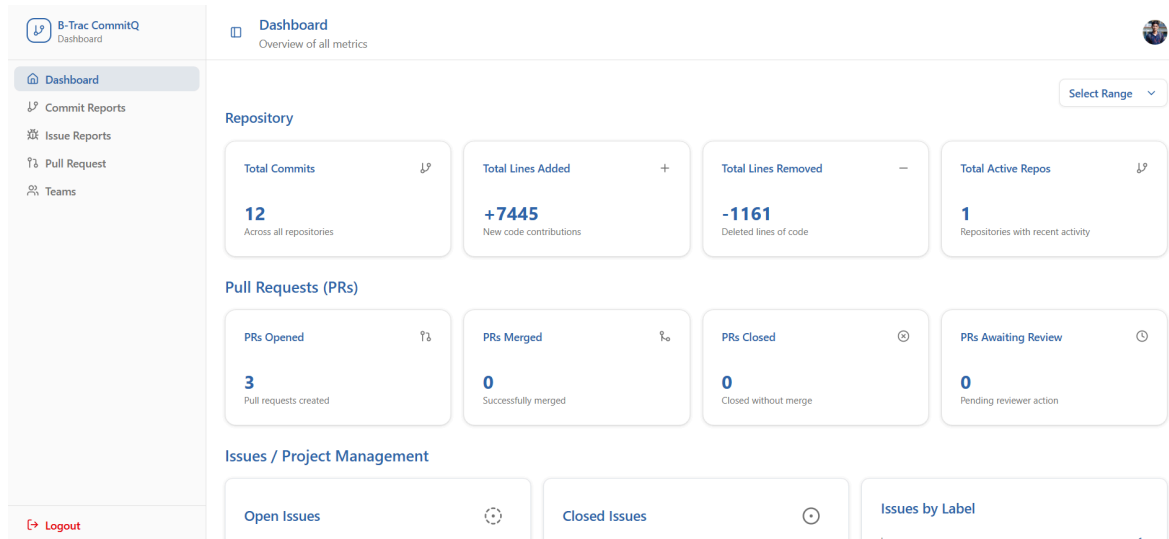


Figure 18: Dashboard of GitHub dashboard

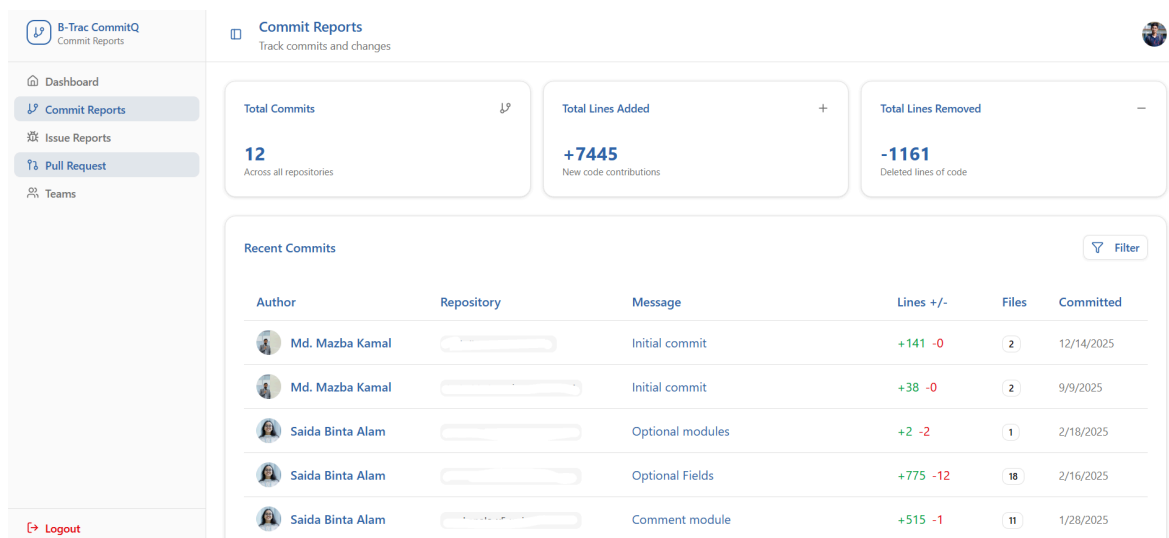


Figure 19: Commit reports of GitHub dashboard



Figure 20: Issue reports of GitHub dashboard

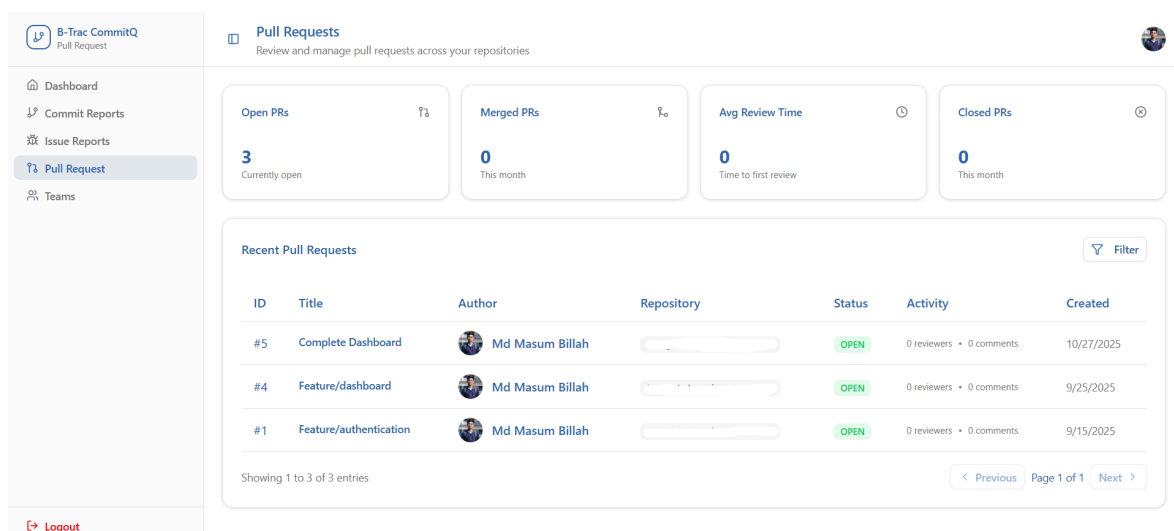


Figure 21: PR reports of GitHub dashboard

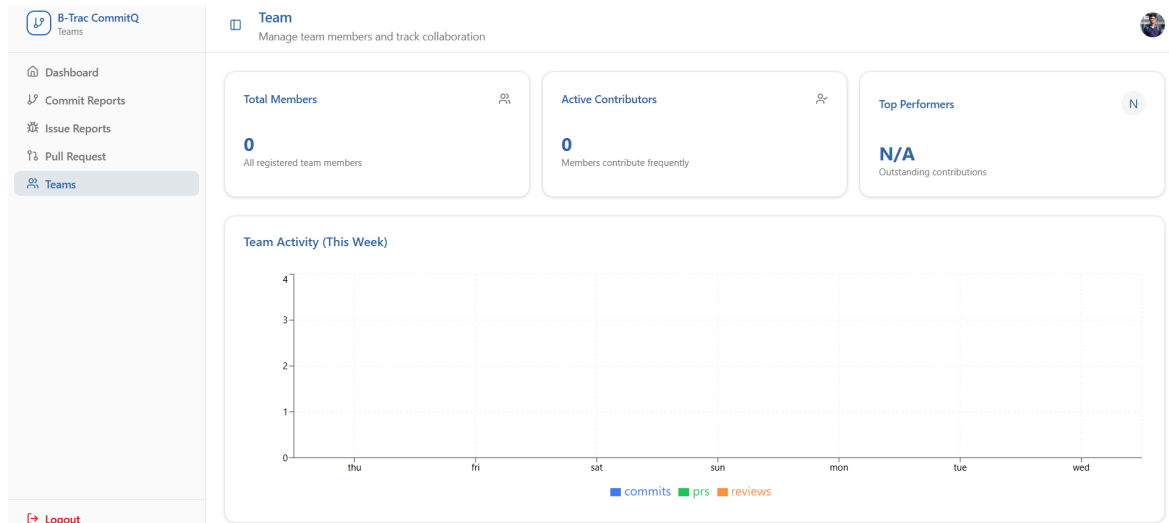


Figure 22: Team analytics of GitHub dashboard

BTSL Disaster Notification

The BTSL Disaster Notification System is a real-time alerting system that I worked on during my internship at B-Trac Solutions Ltd. The primary objective of this system was to deliver timely disaster notifications, specifically earthquake alerts to multiple mobile applications, including Workopolo, Nirapath, Carcopolo, and other related platforms.

The system retrieves earthquake data from the United States Geological Survey (USGS) public API, continuously monitors seismic activity, and determines whether an earthquake has occurred within predefined geographical regions. When a relevant event is detected, the system sends real-time notifications to users across all supported applications.

Features

The key features of the BTSL Disaster Notification System include:

- Tracking earthquake events using the USGS public API based on centralized geographic coordinates
- Periodically polling the public API at 60-second intervals
- Detecting earthquake occurrences within specified regions
- Sending real-time notifications to users
- Broadcasting notifications to all installed devices
- Supporting notification delivery to multiple mobile applications simultaneously

Technologies

- TypeScript
- Node.js
- Commander.js
- Firebase Cloud Messaging (FCM)

Learning

Through the development of this system, I gained hands-on experience in applying software design patterns and software engineering best practices in a real-world scenario. Working with a real-time public API introduced me to challenges related to continuous polling, event detection, and asynchronous processing.

Additionally, implementing notification delivery using Firebase Cloud Messaging enhanced my understanding of real-time communication systems, message broadcasting, and multi-application notification management. This project significantly strengthened my ability to design scalable, reliable, and event-driven backend systems.

Proof of Work

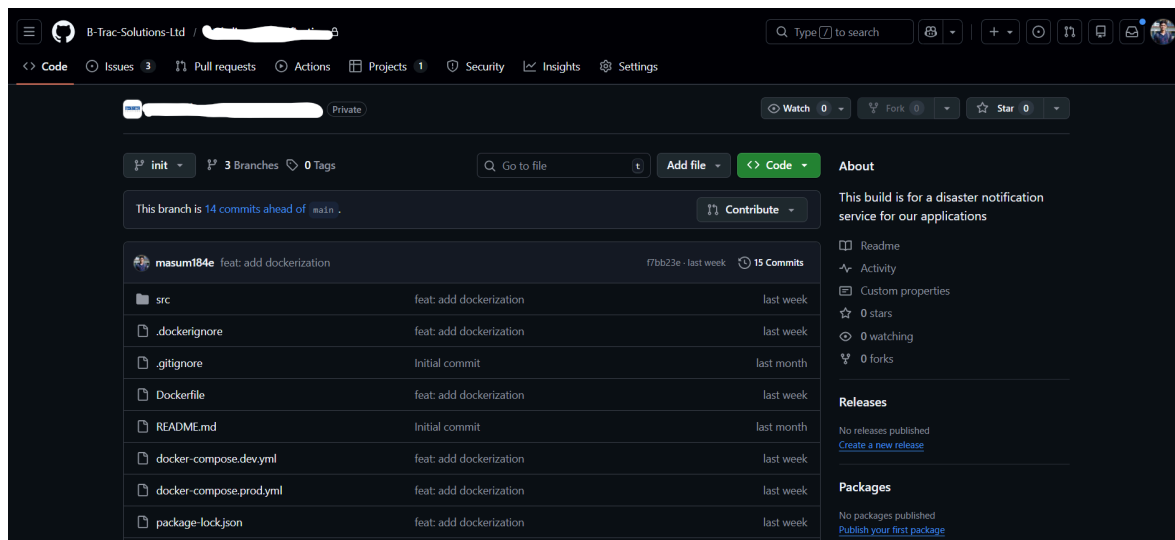


Figure 23: GitHub repository of disaster notification

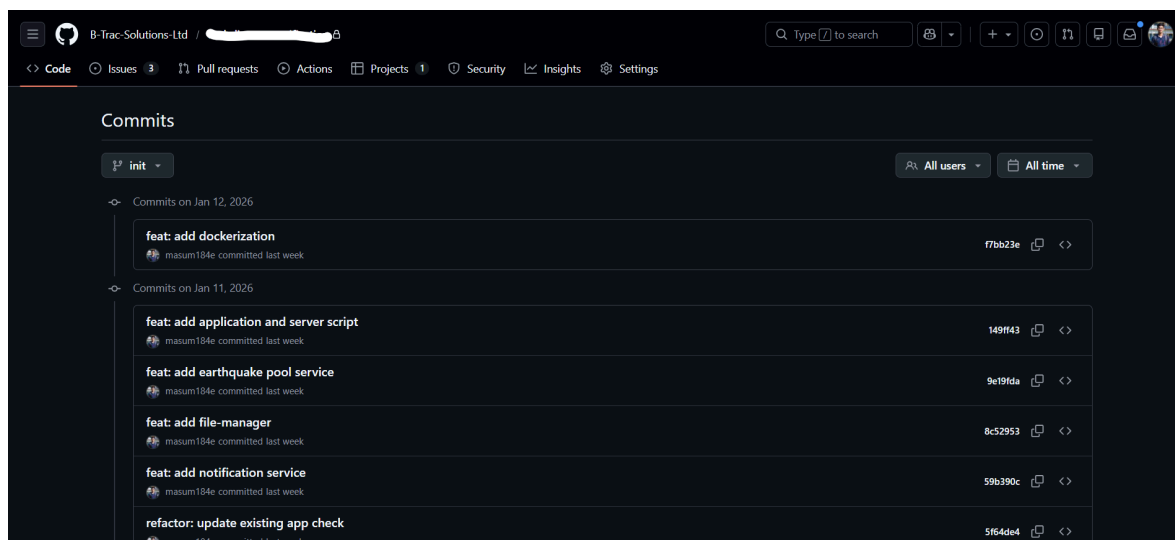


Figure 24: Commit history of disaster notification

My Contributions

Throughout my internship at B-Trac Solutions Ltd, I was actively involved in multiple development initiatives. These contributions allowed me to apply academic knowledge in real-world scenarios, strengthen my software engineering skills, and understand industry-standard development practices.

BTSL GitHub Dashboard

One of my key contributions during the internship was working on the BTSL GitHub Dashboard, an internal analytical tool designed to enhance the productivity of team leaders. My initial responsibility involved gathering and analyzing requirements to identify which types of repository data and metrics would be most valuable for monitoring team performance and development activity.

Based on these requirements, I contributed to the design of a scalable GraphQL API architecture capable of efficiently handling large volumes of GitHub data. The API was structured to support flexible queries, reduce unnecessary data transfer, and ensure optimal performance even when processing data from multiple repositories and contributors. This experience strengthened my understanding of backend design principles, data modeling, and performance-oriented API development.

BTSL Disaster Notification

I contributed to the development of the BTSL Earthquake Disaster Notification System, a real-time mobile-based alert platform designed to disseminate critical earthquake-related information to users in a timely manner. The system was designed following structured software engineering principles to ensure scalability, reliability, and long-term maintainability.

My responsibilities included defining the overall system architecture and implementing the core application logic while adhering to established design pattern principles and industry best practices. Special attention was given to creating a clean and modular codebase to support future enhancements and seamless integration with multiple mobile applications.

In addition, I implemented real-time data acquisition from the USGS public API to retrieve verified earthquake data. The system was designed to process these updates efficiently and deliver real-time notifications to users across multiple installed mobile applications. Through this project, I gained practical experience in real-time system design, external API integration, event-driven communication, and cross-platform notification mechanisms.

Learnings

TypeScript

During my internship, I gained extensive hands-on experience with TypeScript through real-world development in both the BTSL GitHub Dashboard and the BTSL Disaster Notification System. Working in production-level codebases significantly strengthened my understanding of TypeScript's type system, including type inference, custom type definitions, and strict typing for improved reliability. This exposure enhanced my ability to write safer, more maintainable, and scalable code.

Next.JS

While developing the BTSL GitHub Dashboard, I worked extensively with Next.js, which provided me with a deeper understanding of modern React-based frameworks. I gained practical experience in designing user interfaces using shadcn/ui, as well as understanding the architectural distinction between client components and server components.

Additionally, I worked with NextAuth, particularly in managing GitHub Personal Access Tokens. This required storing authentication data securely within the NextAuth state and understanding how internal state modifications affect both the UI layer and browser storage. Through this, I developed a stronger grasp of authentication workflows in full-stack applications.

GitHub GraphQL API

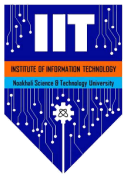
A significant learning outcome was my experience with the GitHub GraphQL API. I studied the differences between GitHub's REST and GraphQL APIs and understood why GraphQL was better suited for the dashboard application due to its flexibility and efficiency. I learned not only how to write optimized GitHub GraphQL queries but also how to structure queries for other public GraphQL-enabled APIs.

Deployments

I gained hands-on experience in deploying applications from scratch to production environments. This included configuring deployment pipelines and resolving real-world issues that arise after deployment. One critical learning experience involved improper URL handling due to a staging server proxy, which initially caused UI failures, broken API calls, and rendering issues. By systematically correcting URL configurations across the codebase, I restored application stability. This experience emphasized the importance of environment-aware development practices.

Real Time Communication

During the development of the BTSL Disaster Notification System, I worked with real-time data processing and communication mechanisms. The system was designed to poll the USGS public earthquake alert API every 60 seconds to detect new earthquake events.



Upon detecting an event, the system sent real-time notifications to all installed mobile applications within the organization. Through this task, I developed a clear understanding of Firebase Cloud Messaging (FCM), including sending notifications to individual devices as well as broadcasting messages to groups of devices.

Design Principles

I had the opportunity to apply software design principles learned during my academic studies to real-world projects. This included implementing modular architectures, separation of concerns, and clean coding practices, which improved the overall quality and maintainability of the systems I worked on.

Command Line Tool

During my internship at B-Trac Solutions Ltd. I was exposed to the process of building and using command-line tools to support development workflows. This experience helped me understand how CLI tools can improve productivity, automate repetitive tasks, and streamline development processes.

Quality of Work

While developing APIs for the BTSL GitHub Dashboard, I encountered limitations with the GitHub GraphQL API when handling large volumes of data. To address this, I designed the API to fetch only the necessary data using efficient pagination strategies, ensuring performance, reliability, and compliance with API limitations. This experience reinforced the importance of designing systems with scalability in mind.

Think Before Code

One of the most important lessons I learned was the value of defining requirements before writing code. During the development of the GitHub Dashboard, the codebase required restructuring multiple times as new ideas emerged through exploration and mentor discussions. This experience taught me that clear requirement specification significantly reduces development time and minimizes rework.

Collaboration

Collaboration played a vital role in my internship experience. While developing the disaster notification system, I needed an Android test application to verify real-time notification delivery. For this, I collaborated with Saiful Islam Sazib (Assistant Manager, Mobile Application). Together, we discussed the requirements and designed a simple test SDK to receive Firebase notifications in real time. This collaboration enhanced my communication skills and highlighted the importance of teamwork in cross-platform development.

Growth

Technical Growth

Throughout my internship, I experienced substantial technical growth by working on real-world, production-level systems. I explored and applied a range of modern technologies, including TypeScript, Next.js, GraphQL, GitHub GraphQL APIs, Firebase Cloud Messaging, and cloud-based deployment workflows.

By contributing to the BTS� GitHub Dashboard and the BTS� Disaster Notification System, I strengthened my understanding of full-stack application development, API design, real-time data processing, and scalable system architecture. I also applied software design principles and best practices learned during my academic coursework, which helped me bridge the gap between theoretical knowledge and practical implementation. Overall, this internship significantly enhanced my problem-solving ability, code quality awareness, and confidence in handling complex technical requirements.

Professional Growth

In addition to technical expertise, the internship contributed greatly to my professional development. Working in a collaborative organizational environment improved my communication skills, teamwork, and ability to interact effectively with mentors, developers, and stakeholders. I learned how to receive feedback constructively, participate in technical discussions, and align my work with organizational goals and timelines.

The internship also enhanced my understanding of professional responsibilities, workplace ethics, and industry expectations. Exposure to structured development processes, deployment cycles, and cross-functional collaboration prepared me for future professional roles and strengthened my readiness to transition from an academic environment to a corporate setting.

Self Assessment

My internship has been a transformative learning experience that contributed significantly to my personal and professional development. Through practical exposure to real-world projects, I gained a clearer understanding of professional responsibilities, workplace dynamics, and industry expectations. Continuous learning, interaction with experienced professionals, and hands-on problem solving enhanced both my technical competence and interpersonal skills. Overall, this internship has prepared me to face future professional challenges with greater confidence, adaptability, and responsibility.

	Outstanding (5)	Very Good (4)	Good (3)	Need Improvements (2)	Poor (1)	Sub-Total
Punctuality Attends office regularly	✓					5
Ability to Solve Problem Ability to identify and resolve the issue	✓					5
Accuracy of Work Correctness in assigned task	✓					5
Creativity Ingenuity shown while solving problems	✓					5
Dedication Completed assigned task within the scheduled time	✓					5
Professionalism Attitude at work environment	✓					5
Teamwork Ability to collaborate with teammember		✓				5
Interpersonal Communication Maintain two way effective communication	✓					4

Leadership Take initiative and contribute ideas to improve outcome		✓				4
	✓					5
Total Count (Out of 50)						48

In the above self-assessment, I have evaluated my performance based on the skills and competencies I developed during my internship period. I consistently attempted to perform my duties with sincerity, responsibility, and professionalism. While I achieved strong results in most areas, I recognize that there is still room for improvement, particularly in leadership and advanced communication skills. Considering my overall performance and learning progress, I would rate myself 48 out of 50, acknowledging both my achievements and areas for future growth.

Time Management

Throughout the internship, I developed strong time management skills by balancing assigned tasks, learning new technologies, and meeting project deadlines. I learned to prioritize work based on urgency and impact, which helped me deliver features within expected timelines while maintaining quality.

Ability to learn

The internship environment required continuous learning due to the use of new tools, frameworks, and APIs. I adapted quickly by researching independently, seeking guidance from mentors when necessary, and applying newly acquired knowledge directly to ongoing projects. This experience strengthened my ability to learn efficiently in a professional setting.

Accuracy of work

I maintained a strong focus on accuracy by carefully analyzing requirements, validating outputs, and testing my implementations thoroughly. Attention to detail became essential, particularly when working with APIs, real-time data, and production-level code, where minor errors could lead to significant issues.

Creativity

The internship encouraged me to think creatively when addressing technical challenges. I explored alternative solutions, optimized existing approaches, and contributed ideas during discussions with mentors. This helped me enhance my problem-solving mindset while aligning creativity with practical constraints.

Dedication and Work Ethic

I demonstrated dedication by taking ownership of my responsibilities and maintaining consistency in my efforts. I remained proactive in completing tasks, improving my understanding of the systems, and contributing meaningfully to the projects assigned to me.

Professional Conduct

Working in a corporate environment improved my professionalism in terms of communication, responsibility, and adherence to organizational standards. I learned to respect deadlines, follow proper documentation practices, and maintain a professional attitude during meetings and collaborative discussions.

Team Work and Collaboration

The internship provided valuable exposure to teamwork, where coordination with mentors and fellow developers was essential. I actively participated in discussions, shared progress updates, and collaborated effectively to achieve shared project goals.

Adaptation of New Technology

I significantly enhanced my ability to adopt new technologies by working with modern frameworks and tools that were previously unfamiliar to me. This experience improved my confidence in exploring, understanding, and applying emerging technologies in real-world projects.

Interpersonal Communication

Regular interaction with mentors and team members improved my interpersonal communication skills. I learned how to express ideas clearly, ask relevant questions, and provide updates in a structured and professional manner.

Leadership

Although I was an intern, I demonstrated leadership qualities by taking initiative in problem identification, proposing solutions, and independently managing assigned tasks. This experience helped me understand the importance of responsibility, accountability, and initiative in a professional environment.

Conclusion

This academic internship at B-Trac Solutions Ltd. was a transformative experience that played a critical role in shaping my technical capabilities and professional mindset. It provided me with a platform to apply academic learning in real-world scenarios, gain hands-on exposure to modern technologies, and understand the practical challenges of software development.

The knowledge, skills, and professional values acquired during this internship have laid a strong foundation for my future career. This experience not only enhanced my technical competence but also clarified my career goals and motivated me to pursue continuous learning and growth in the software engineering domain.